E-commerce sales & profitability Analysis

1. This project analyzes sales performance, profitability trends, and customer behavior for a large e-commerce retailer.

2. The objective is to identify revenue drivers, evaluate the impact of discounting strategies, and uncover opportunities to improve profitability through data-driven insights and strategic recommendations.

Business problems

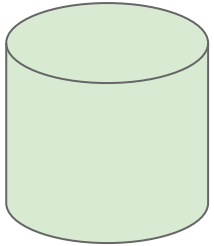
Revenue growth slowing,Profit margins dropping.Customer purchasing becoming inconsistent.

- ❑ Which regions, cities, and categories drive revenue?
- ❑ Where is profit leaking?
- ❑ Which customer segments are most valuable?
- ❑ How can we improve sales and profitability?

Data overview

--------**Dataset size**

- ➔ 50000 transactions
- ➔ 10000 customers
- ➔ 2 years of data



Data Structure

1. 15 columns
2. Transaction-levels dataset.

Key Fields - order id , customer id,category,Revenue,Profit,Discount rate

Key KPIs

Total Profit Rs- 2.51B

Profit Margin % - 17%

Unique Customers - 9924

Average Order Value (AOV) - Rs50.25K

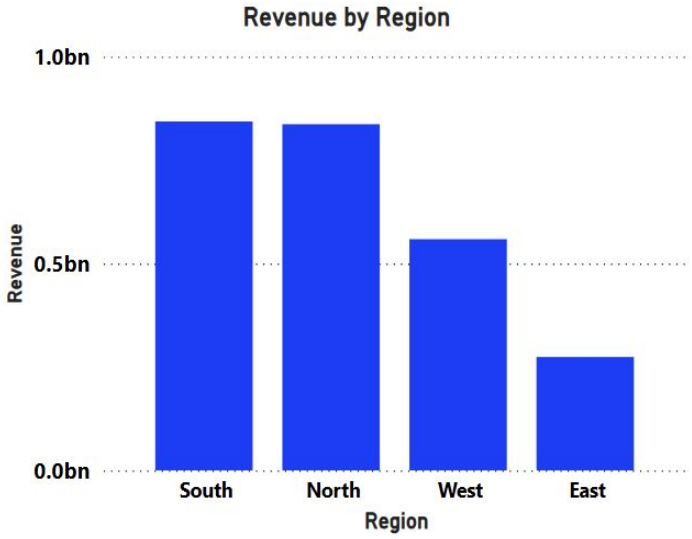
Orders per Customer - 5 OPC

Revenue per Customer - Rs 253.16K

Average discount - 20%

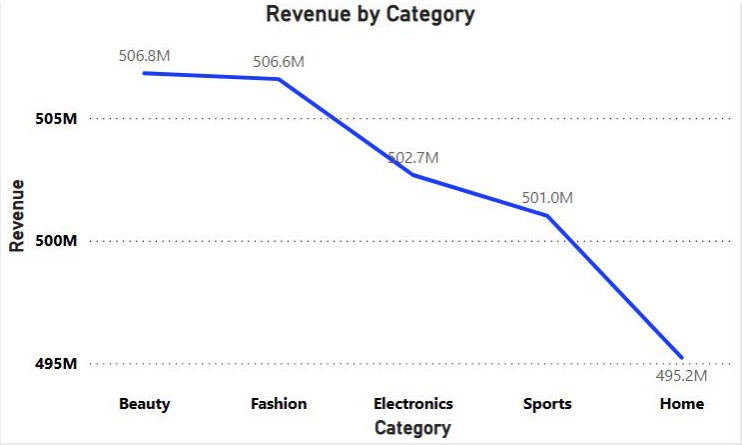
Revenue analysis

1. North region contributes 38% of total Revenue.
2. South region shoes fastest growth.
3. East region has lowest sales performance.



Category Performance

1. The top performing category is Beauty , Fashion with 506.8M .
2. Lowest performing category is Home With 495.2M .



City performance

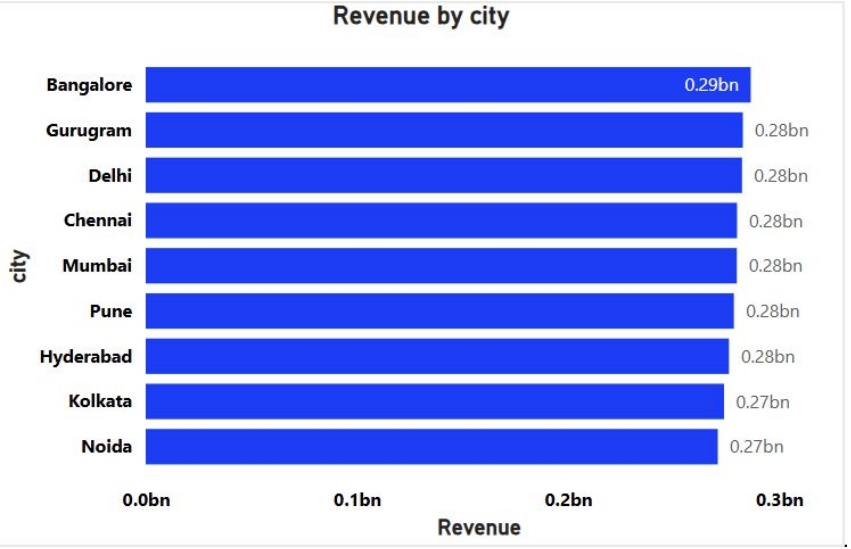
Bangalore

Gurugram

Delhi

Chennai

Mumbai



Customer analysis

1. The majority of customers fall in the 4–10 orders segment, indicating a strong base of repeat purchasers.
2. These 10+ order customers are power customer and these are repeater.

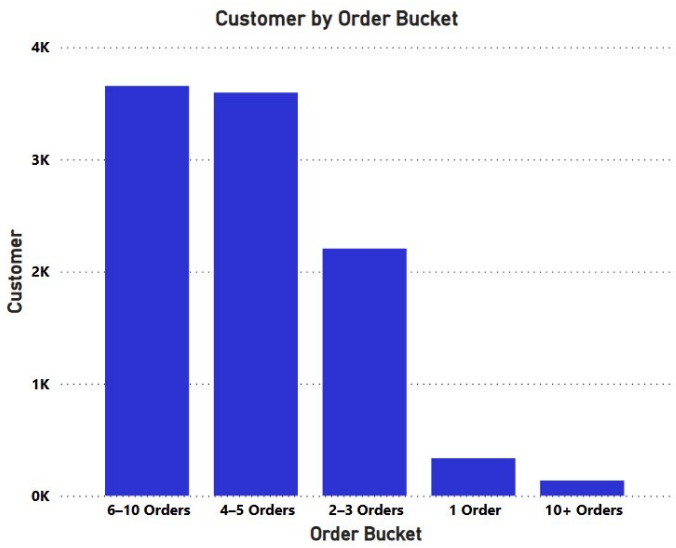
KPIs

Revenue per customer-253.16K

Average order value(AOV)-50.25K

Unique customers-9924

Order per customer(OCP)-5

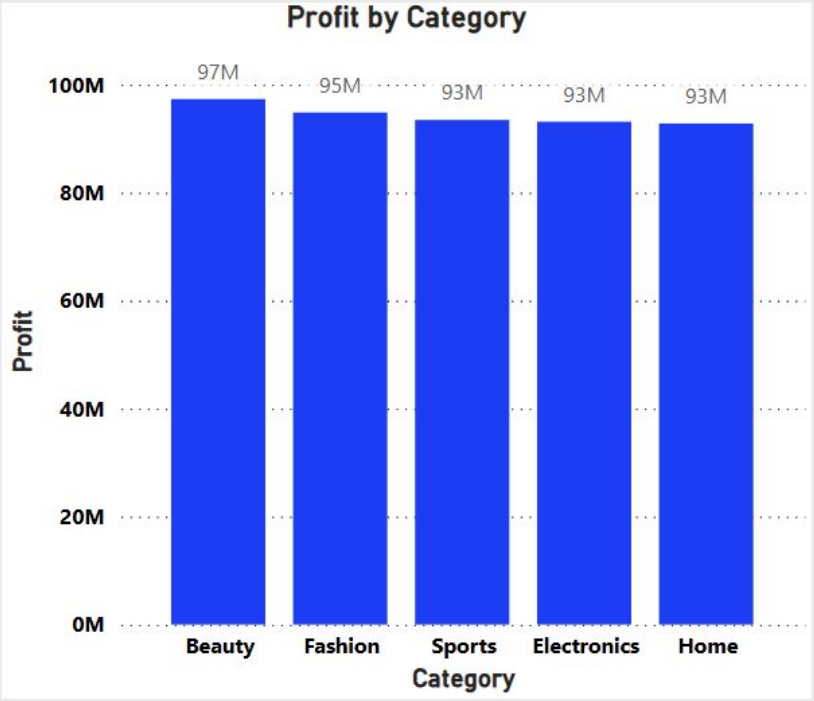


Profitability Problem

1. Beauty category generates the highest profit (97M), indicating Strong margins and customer demand compared to other categories. Fashion follows closely.

2. Sports, Electronics, and Home show identical profits (~93M) indicating relatively stable but slightly lower profitability across these categories.

3. Profit distribution across categories is relatively balanced, with only a 4M gap between the highest and lowest, suggesting no major underperforming category.



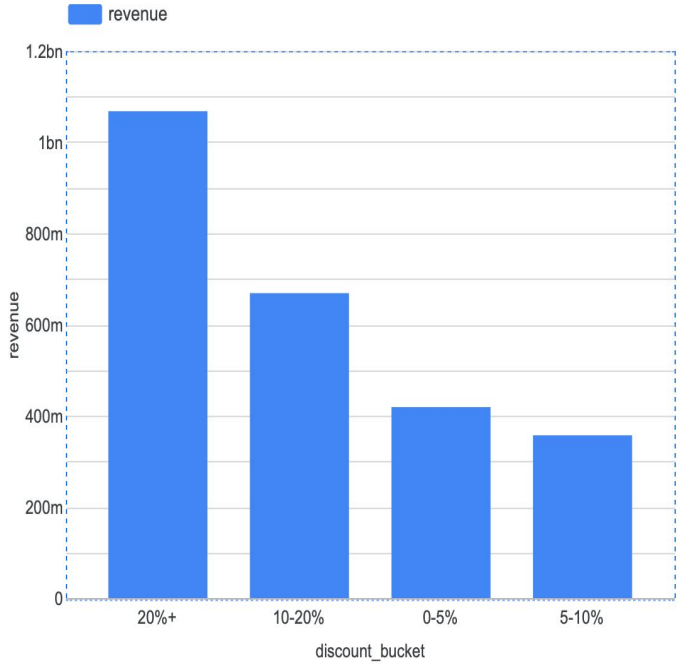
Discount impact

1. Deep discounts drive the highest revenue contribution .The 20%+ discount bucket generates the highest revenue (~1.07B), significantly outperforming all other discount ranges, indicating that aggressive promotions strongly drive overall sales volume.

2. Revenue declines steadily with lower discounts revenue drops from ~670M in the 10–20% range to ~420M and ~360M in the 0–5% and 5–10% ranges, suggesting that smaller discounts are less effective at stimulating customer spending.

3. Revenue increases sharply when discounts exceed 20%, indicating strong customer price sensitivity at this threshold.

4. Revenue growth may come at the cost of margins While 20%+ discounts generate the highest revenue, heavy reliance on deep discounting could reduce profit margins, highlighting the need to balance volume growth with profitability.



Recommendations

1. Optimize Discount Strategy
Average discounts are high (20%+) while margins remain very low. Shift from blanket discounts to targeted or loyalty-based promotions to reduce discount levels while maintaining demand.

2. Focus on High-Performing Regions
Revenue and profitability are strongest in the North and South regions. Increase marketing and operational investment in these regions while reviewing the cost structure of the East region.

3. Strengthen High-Value Customer Retention
Most customers fall in the 4–10 order range, with a sharp drop beyond 10 orders. Launch a VIP or loyalty program to move these customers into the 10+ order segment and increase revenue per customer.

4. Improve Underperforming Categories
While Beauty drives the highest revenue and profit, the Home category underperforms. Optimize pricing, reduce supply costs, and reallocate inventory toward higher-demand categories like Electronics and Sports.

Conclusion

Conclusion: Key Takeaways

- **Revenue heavily depends on high discount levels.**
- **Profit margins remain low due to aggressive promotions.**
- **Revenue and profit are concentrated in specific regions and categories.**
- **Customer retention presents a strong opportunity for future growth.**